

Building an **Air Quality Monitoring System** moves you into "Medium" complexity by introducing gas sensing and advanced data visualization. This project is essential for smart city applications and health monitoring.

Task 5 — Project Plan

- **Duration:** 8 hours.
- **Complexity:** Medium.
- **Goal:** Monitor air quality using gas sensors and store/visualize the results.

Required Resources

- **Sensors:** MQ-135 (Air Quality Sensor for detecting NH₃, NO_x, alcohol, Benzene, smoke, and CO₂).
- **Microcontroller:** Arduino or Raspberry Pi.
- **Output:** LCD Display (for real-time viewing) or an SD card module (for storage).
- **Buzzer (Optional):** To create an alert system if air quality drops below a safe threshold.

Technical Implementation

1. Hardware Connections

The **MQ-135** sensor has both Digital and Analog outputs. For precise monitoring, we use the **Analog (A0)** pin to get varying PPM (parts per million) values.

MQ-135 Pin	Arduino Pin
VCC	5V
GND	GND
A0 (Analog Out)	A0

2. Core Programming Logic

The sensor requires a "pre-heat" time to provide stable readings. The code reads the analog voltage, converts it to a representative air quality value, and outputs it to your storage or display.

```
int sensorPin = A0;
int sensorValue;

void setup() {
  Serial.begin(9600);
  pinMode(sensorPin, INPUT);
}

void loop() {
  sensorValue = analogRead(sensorPin); // Read air quality

  Serial.print("Air Quality (PPM): ");
  Serial.println(sensorValue);
}
```

```
// Storage or Display logic here  
delay(1000);  
}
```

Learning Outcomes

- **Air Quality Measurement:** Understanding how chemical sensors interact with the atmosphere to produce electrical signals.
- **Data Visualization:** Converting raw sensor data into meaningful charts or real-time alerts to track environmental health over time.

Real-World Application

This system is a foundational block for **Smart City Health Monitoring**, allowing local governments or individuals to track pollution levels in specific neighborhoods or rooms.